## A. SCHIFF MOMENT

Does the fact that TlF is a molecule, not an atom, change the following derivation?

**Shielding potential** Following appendix of ref. [22], I find an expression that takes care of shielding nuclear EDM effects as per the Schiff theorem by adding a term with zero expectation value to an atomic Hamiltonian.

Consider an atom with electrons at positions $\mathbf{r}_i$ and momenta $\mathbf{p}_i$ and a nucleus with a charge distribution $e\rho(\mathbf{r})$ and an electric dipole moment $\mathbf{d}$ in an external electric field $\mathbf{E}_0$ having the following terms in its Hamiltonian:

$$\hat{H} = \sum_i \left[\frac{\mathbf{p}_i^2}{2m} - e\phi_0(\mathbf{r}_i) - e\mathbf{r}_i \cdot \mathbf{E}_0\right] + \phi_1 - \mathbf{d}\cdot\mathbf{E}_0,$$

$$\phi_0 \equiv e\int \frac{\rho(\mathbf{r}')}{|\mathbf{r}_i - \mathbf{r}'|}\mathrm{d}^3\mathbf{r}', \quad \phi_1 \equiv \sum_{i>k}\frac{e^2}{|\mathbf{r}_i - \mathbf{r}_k|}.$$

The sum over $i$ is taken over all the electrons in the atom, and contains their kinetic energy $\mathbf{p}_i^2/2m$, their energy in the electrostatic potential $\phi_0$ of the nucleus and in the external field. In addition, we include the electron–electron electrostatic interaction $\phi_1$, and the energy of the nuclear dipole $\mathbf{d}$ in the external field.

Let $|n\rangle$ be an eigenstate of the atomic Hamiltonian $\hat{H}$ with energy $E_n$:

$$\hat{H}|n\rangle = E_n|n\rangle.$$

Then the commutator of an arbitrary operator $\hat{O}$ with $\hat{H}$ has a zero expectation value in that state:

$$\begin{aligned}\langle n|[\hat{H},\hat{O}]|n\rangle &= \langle n|\hat{H}\hat{O}|n\rangle - \langle n|\hat{O}\hat{H}|n\rangle \\ &= E_n\langle n|\hat{O}|n\rangle - \langle n|\hat{O}|n\rangle E_n \\ &= 0.\end{aligned}$$

Define the perturbation $V$ as the commutator of the total electron momentum with the atomic Hamiltonian, multiplied by $\mathbf{d}/ie\hbar Z$:

$$\begin{aligned}ie\hbar ZV &\equiv \mathbf{d}\cdot\left[\sum_i \mathbf{p}_i,\ \hat{H}\right] \\ &= \mathbf{d}\cdot\left[\sum_i \mathbf{p}_i,\ \sum_j\big(-e\phi_0(\mathbf{r}_j) - e\mathbf{r}_j\cdot\mathbf{E}_0\big)\right],\end{aligned}$$

noting that the sum of momenta trivially commutes with the electron kinetic energy, the electron–electron interaction energy (why that one?), and the energy of the nuclear dipole moment in the external field. The canonical commutation relation $[\mathbf{r}_i, \mathbf{p}_j] = i\hbar\delta_{ij}$ allows us to transform the above into

$$i\hbar ZV = -\mathbf{d}\cdot\sum_i \underbrace{[\mathbf{p}_i,\ \phi_0(\mathbf{r}_i)]}_{-i\hbar\nabla_i\phi_0(\mathbf{r}_i)} - \mathbf{d}\cdot\underbrace{\sum_i[\mathbf{p}_i,\ \mathbf{r}_i]}_{-i\hbar Z}\mathbf{E}_0,$$

where the fact that $[\mathbf{r}_i,\ [\mathbf{r}_i, \mathbf{p}_i]] = 0$ allows taking the $\phi_0$ outside the commutator and differentiating it. Cancelling common terms, we are left with the expression

$$V = \mathbf{d}\cdot\mathbf{E}_0 - \frac{1}{Z}\sum_i \mathbf{d}\cdot\nabla_i\phi_0(\mathbf{r}_i).$$

How did the above two terms end up with a different sign?

Since $V$ has been derived from a commutator of an operator with the Hamiltonian $\hat{H}$, it has a vanishing expectation value in any superposition of eigenstates of $\hat{H}$ as shown above and it does not change the energy levels to first order in perturbation theory. We are thus at liberty to add such a term to the Hamiltonian: it should not change the result of an atomic EDM measurement, which looks for an energy shift linear in the external field.

Adding $V$ to the Hamiltonian, we see that the energy $-\mathbf{d}\cdot\mathbf{E}_0$ of the nuclear dipole in the external field disappears. In addition, the electrons–nucleus electrostatic potential $\phi_0(\mathbf{r}_i)$ is modified to

$$e\int\frac{\rho(\mathbf{r}')}{|\mathbf{r}_i - \mathbf{r}'|}\mathrm{d}^3\mathbf{r}' + \frac{1}{Z}(\mathbf{d}\cdot\nabla_i)\int\frac{\rho(\mathbf{r}')}{|\mathbf{r}_i - \mathbf{r}'|}\mathrm{d}^3\mathbf{r}', \tag{12}$$

which matches eqn 5 if we drop the indices $i$.

We have been allowed to add $V$ *only* because to first order it does not change the energy levels. We now proceed to show that adding $V$ did, after all, change the energies—the dipole terms cancel, and Schiff moment appears. That sounds really shady: $V$ was added on the premise that it makes no difference, and yet we're now trying to prove that it does make a difference???

**Multipole expansion** The dipole cancellation can be seen via the multipole expansion [56, eqn 3.38]

$$\frac{1}{|\mathbf{r} - \mathbf{r}'|} = \sum_{\ell=0}^{\infty}\frac{r_<^\ell}{r_>^{\ell+1}}P_\ell(\cos\gamma),$$

$P_\ell$ being the Legendre polynomials of degree $\ell$, and $\gamma$ the angle between $\mathbf{r}$ and $\mathbf{r}'$. We use the notation $r_<, r_>$ to denote the smaller/greater of $r$ and $r'$.

Following [70, sect II], we can use this to expand the modified potential from eqn 12. The first term becomes, taking just the $\ell = 1$ term,

$$e\int\rho(\mathbf{r}')\mathrm{d}^3\mathbf{r}'\left[\frac{r'}{r^2}(r' < r) + \frac{r}{(r')^2}(r' > r)\right]P_1(\cos\gamma), \tag{13}$$

using Boolean comparison expression $(r' > r)$ that evaluates to 1 if the condition is satisfied, and to 0 otherwise.

Similarly, the $l = 0$ part of the second term in eqn 12 is

$$\frac{1}{Z}(\mathbf{d}\cdot\nabla)\int\rho(\mathbf{r}')\mathrm{d}^3\mathbf{r}'\left[\frac{1}{r}(r' < r) + \frac{1}{r'}(r' > r)\right]P_1(\cos\gamma). \tag{14}$$

Noting that the nuclear EDM and the nuclear charge are defined as

$$\mathbf{d} \equiv e \int \mathbf{r}' \rho(\mathbf{r}') \mathrm{d}^3\mathbf{r}, \quad Z = \int \rho(\mathbf{r}') \mathrm{d}^3\mathbf{r}',$$

the expression 14 becomes

$$\frac{1}{Z} \nabla \cdot \int_0^\infty e\mathbf{r}' \rho(\mathbf{r}') \mathrm{d}^3\mathbf{r}' \int_0^r \frac{1}{r} \rho(\mathbf{r}') \mathrm{d}^3\mathbf{r}',$$

noting that $\nabla(1/r') = 0$ since $\nabla$ only acts on the non-primed coordinates. That also allows us to move the $\nabla$ into the first integral, and move the $1/r$ from the second integral to the first:

$$\frac{e}{Z} \int_0^\infty \underbrace{\mathbf{r}' \cdot \nabla \left( \frac{1}{r} \right)}_{-r'/r^2 \cos\gamma} \rho(\mathbf{r}') \mathrm{d}^3\mathbf{r}' \underbrace{\int_0^r \rho(\mathbf{r}') \mathrm{d}^3\mathbf{r}'}_{Z \text{ when } r \to \infty}$$

In the limit $r \to \infty$, this exactly cancels the first term in eqn 13, thus screening all nuclear EDM. Is that a correct conclusion? There is still the second term in eqn 13, why does that not need screening? Or is that precisely the origin of the Schiff moment? The dipole defunct, we search for the next $P, T$-odd term in the multipole expansion. We just showed how the expression cancels itself out and screens nuclear EDM, and now we proceed to show that there is, after all, some EDM remaining???

**Residual EDM** The two terms that cancel in the limit $r \to \infty$ effectively have $\int_0^\infty = 0$, and so we can change their limits of integration as $\int_0^r \to -\int_r^\infty$. The second term in eqn 13 already has these limits, so we can put all three terms together into the expression

$$\phi^{(1)}(\mathbf{r}) \equiv e\mathbf{r} \cdot \int_r^\infty \left( \frac{\langle \mathbf{r} \rangle}{Zr^3} - \frac{\mathbf{r}'}{r^3} + \frac{\mathbf{r}'}{(r')^3} \right) \rho(\mathbf{r}') \mathrm{d}^3\mathbf{r}',$$

making use of the notation $\langle \mathbf{r} \rangle \equiv \int \mathbf{r}' \rho(\mathbf{r}') \mathrm{d}^3\mathbf{r}'$. Let's evaluate the matrix elements of this potential between s and p Dirac orbitals

$$W_{\mathrm{sp}} \equiv \int \psi_{\mathrm{s}}^*(\mathbf{r}) \phi^{(1)}(\mathbf{r}) \psi_{\mathrm{p}}(\mathbf{r}) \mathrm{d}^3\mathbf{r}.$$

Split off the angular part of the integral:

$$\begin{aligned} W_{\mathrm{sp}} = e \int \psi_{\mathrm{s}}^*(\mathbf{r}) \hat{\mathbf{r}} \psi_{\mathrm{p}}(\mathbf{r}) \sin\theta \mathrm{d}\theta \mathrm{d}\phi \\ \times \int_0^\infty r \underbrace{(\psi_{\mathrm{s}}^* \psi_{\mathrm{p}})}_{\propto r} r^2 \mathrm{d}r \int_r^\infty (\cdots) \rho(\mathbf{r}') \mathrm{d}^3\mathbf{r}', \end{aligned}$$

where $\hat{\mathbf{r}} = \mathbf{r}/r$. In the nonrelativistic case, $\psi_{\mathrm{s}}^* \psi_{\mathrm{p}} \propto r$ as per [70, appendix], while in the radial part we can swap the order of integration $\mathbf{r} \leftrightarrow \mathbf{r}'$:

$$\begin{aligned} W_{\mathrm{sp}} &\propto e \int_0^\infty \rho(\mathbf{r}') \mathrm{d}^3\mathbf{r}' \int_0^{r'} r^2 \left( \frac{\langle \mathbf{r} \rangle}{Zr^3} - \frac{\mathbf{r}'}{r^3} + \frac{\mathbf{r}'}{(r')^3} \right) r^2 \mathrm{d}r \\ &= e \int \rho(\mathbf{r}') \mathrm{d}^3\mathbf{r}' \left( \frac{\langle \mathbf{r} \rangle (r')^2}{2Z} - \frac{\mathbf{r}'(r')^2}{2} + \frac{\mathbf{r}'(r')^5}{5(r')^3} \right) \\ &\propto \frac{e}{10} \left[ \langle r^2 \mathbf{r} \rangle - \frac{5}{3Z} \langle r^2 \rangle \langle \mathbf{r} \rangle \right] \end{aligned}$$

The last line is exactly the expression for the Schiff moment stated in eqn 6. Where did I lose the $4\pi\nabla\delta^3(\mathbf{r})$ factor? Did it go into the angular integral?